

Module 02: Agents — The Counter

Learning Objectives

1. Understand that **YOU** are an Agent who can Count.
2. Learn that Numbers are tools to help Agents reach **Goals**.
3. Know that counting helps us solve mysteries!

Introduction: The Case of the Missing Cookies

Detective Brain has a mystery. There were **5** cookies in the jar. Now there are only **2**. What happened?

- **The Agent:** Detective Brain.
- **The Tool:** Counting (Math).
- **The Goal:** Find out who ate 3 cookies!

An Agent uses numbers to understand the world. Without numbers, we wouldn't know any were missing!

Key Concepts

1. The Counter

You are a "Counter." When you point and say "One, Two, Three," you are doing a job. You are turning a pile of things into a **Number**.

2. One-to-One Correspondence

This is a big rule for Agents. You must touch **One Item** for **One Number**.

- Touch Cookie -> Say "One."
- Touch Cookie -> Say "Two." If you say "One, Two" but only touch one cookie, you are tricking yourself!

3. Cardinality (The Last Number)

When you count "1, 2, 3, 4, 5"... The last number (5) is the magic number. It tells you "How Many in the whole pile!"

Activities

Activity 1: The Detective Walk

Walk around your room. Count things!

- How many windows?
- How many chairs?
- How many doors? Write it down in your Detective Notebook.

Activity 2: The Cookie Mystery

Get 5 toys (cookies). Close your eyes. Ask a friend to take some away. Open your eyes. Count them. How many are left? How many are missing? ($5 - ? = ?$) You are doing subtraction!

Summary

Agents use counting to measure the world. By following the rules (One-to-One), we can find out "How Many" and solve mysteries like the Missing Cookies!

References

- *The Doorbell Rang* by Pat Hutchins (Sharing cookies!)
- *Ten Black Dots* by Donald Crews